



N.B.:

1. Figures shown in the right margin indicate full marks
2. Answer Six questions, taking Three from each section including Q.1, Q.2, Q.5 and Q.6
3. Use separate answer script for each section.
4. Assume reasonable data if necessary.

Section-A

- Q.1 (a)** For the **Figure: Q.1(a)**, sketch a free body of the members designated by the letter A and B. All plane surfaces and pins are smooth (frictionless). Flexible, weightless cords are represented by single lines. [CLO1, PLO1, C2] 4

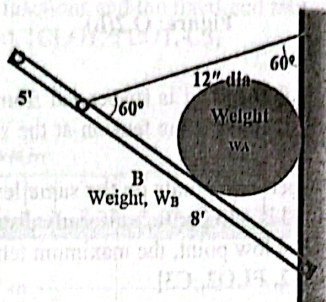


Figure: Q.1(a)

- (b)** A light string ABCDE whose extremity A is fixed, has weights W_1 and W_2 attached to it at B and C. It passes round a small smooth peg at D carrying a weight of 300 N at the free end E as shown in **Figure: Q.1(b)**. If in the equilibrium position, BC is horizontal and AB and CD make 150° and 120° with BC, find (i) Tensions in the portion AB, BC and CD of the string and (ii) Magnitudes of W_1 and W_2 . [CLO1, PLO1, C3] 6

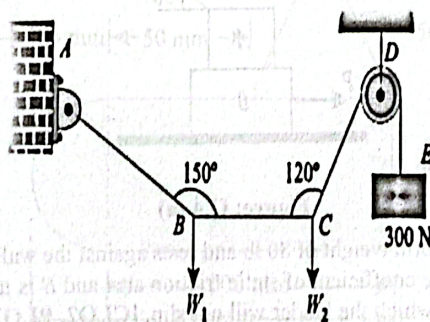


Figure: Q.1(b)

- Q.2 (a)** A sphere "A" of weight 100 N moves on the x-axis in its positive direction with a velocity of 4 m/s. Another sphere "B" of weight 80 N, moving on the same axis but in the opposite direction, approaches it with a velocity of 6 m/s and impinges on it as shown in **Figure: Q.2(a)**. If the coefficient of restitution is 0.7. Calculate the velocities of the two spheres after impact. Also, calculate the loss of kinetic energy that takes place. [CLO3, PLO2, C3] 5

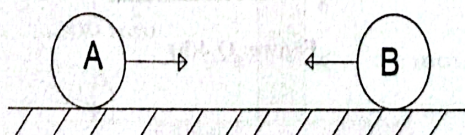


Figure: Q.2(a)

- (b)** A bullet of mass 0.01 kg moving with a velocity of 100 m/s and hit a ball A of a simple pendulum horizontally as shown in **Figure: Q.2(b)**. Find the maximum angle θ through which the pendulum swing when, 5

- i. The bullet gets embedded in the ball.
- ii. The bullet rebounded from the surface with a velocity of 20 m/s.
- iii. The bullet escapes from the other end with a velocity of 20 m/s. Assume mass of the ball to be 1 kg and length of the pendulum to be 1 meter. [CLO3, PLO2, C3]

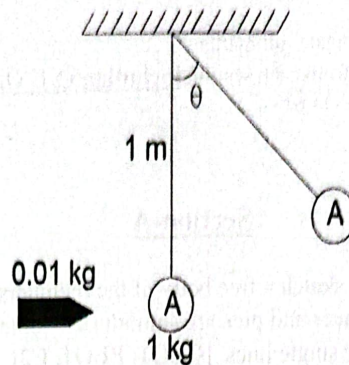


Figure: Q.2(b)

- Q.3 (a) A 80 ft (2S) cable weighs 460 lb and is suspended from two points at the same elevation. For a sag of 20 ft, find (a) the tension at the supports and (b) the span length. [CLO2, PLO2, C3] 5
- (b) A cable is suspended between two points on the same level with a span of 800 ft and a sag of 70 ft. If the load is 4 lb per ft., uniformly distributed horizontally, find the tension in the cable at the low point, the maximum tension and the slope of the cable at the supports. [CLO2, PLO2, C3] 5
- Q.4 (a) Two blocks A and B of weights 1 kN and 2kN respectively are in equilibrium position as shown in Figure: Q.4(a). If the coefficient of friction between the two blocks as well as the block B and the floor is 0.3, find the force P required to move the block B. [CLO2, PLO2, C3] 5

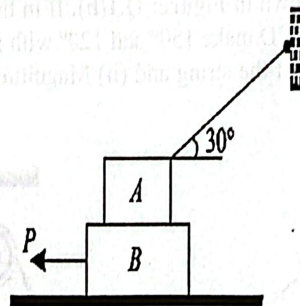


Figure: Q.4 (a)

- (b) The ladder has a uniform weight of 80 lb and rests against the wall at B as shown in Figure: Q.4(b). If the coefficient of static friction at A and B is $\mu = 0.4$, determine the smallest angle at which the ladder will not slip. [CLO2, PLO2, C3] 5

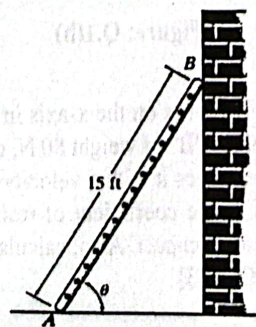


Figure: Q.4(b)

Section-B

- Q.5 (a) Calculate the support reactions for the beam shown in Figure: Q.5(a). [CLO1, PLO1, C3] 5

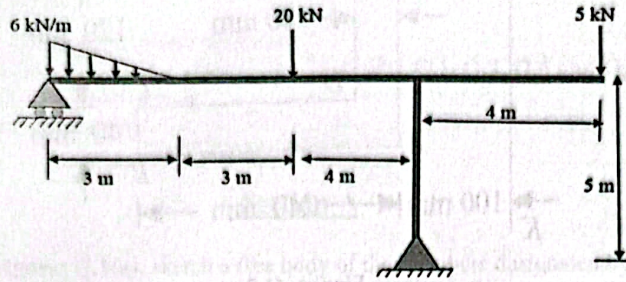


Figure: Q.5(a)

- (b) Determine the support reactions and the fixed-end moment at C for the beam shown in Figure Q.5(b). [CLO1, PLO1, C3] 5

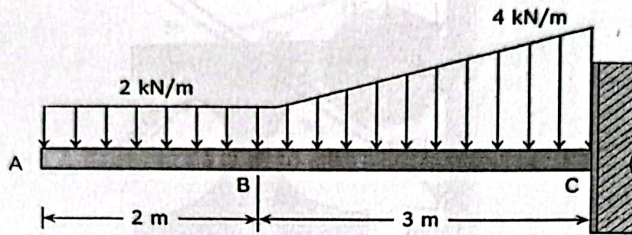


Figure Q.5(b)

- Q.6 (a) The uniform lamina shown in Figure: Q.6(a) consists of a rectangle, a circle, and a triangle. Determine the position of its center of gravity. [CLO2, PLO2, C3] 5

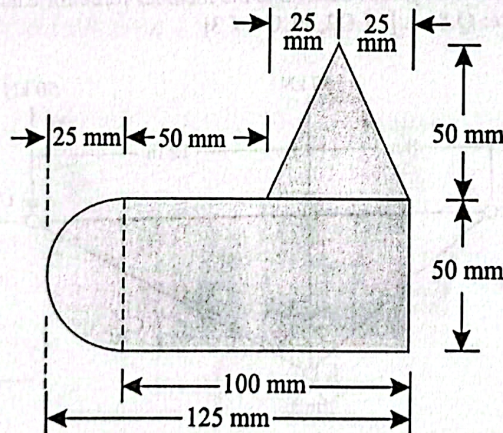


Figure: Q.6(a)

- (b) Determine the centroid of the composite area as shown in Figure: Q.6(b). [CLO2, PLO2, C3] 5

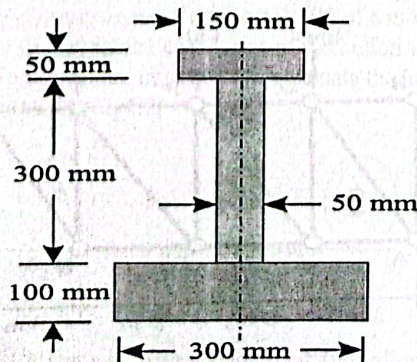


Figure: Q.6 (b)

- Q.7 (a) Find the moment of inertia of the L-section area about the K-K axis, as shown in Figure: Q.7(a). [CLO2, PLO2, C3]

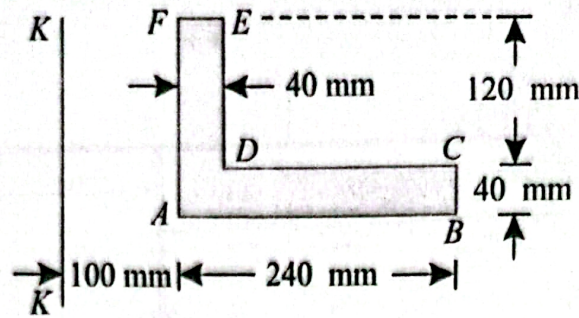


Figure: Q.7(a)

- (b) Determine the moments of inertia of the section about the horizontal and vertical axes passing through the centroid of the section, as shown in Figure: Q.7(b). [CLO2, PLO2, C3]

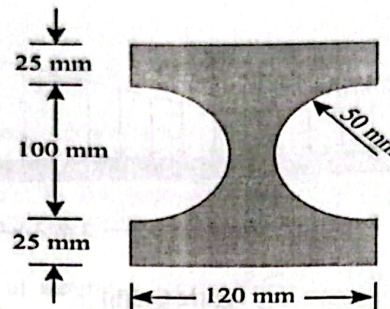


Figure: Q.7(b)

- Q.8 (a) By using the method of joints, determine the member force for each member as shown in Figure: Q.8(a). [CLO2, PLO2, C3]

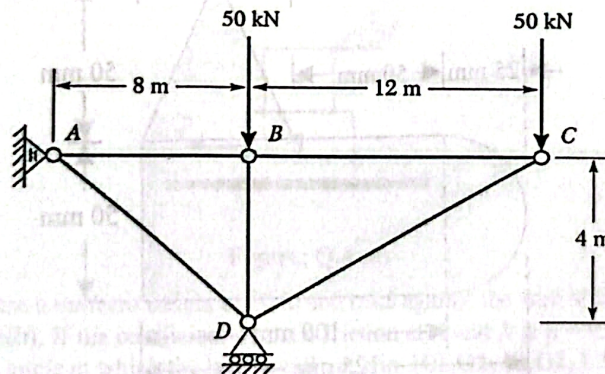


Figure: Q.8(a)

- (b) By using the method of sections, determine the member force of DF, CF, and CE for the member as shown in Figure: Q.8(b). [CLO2, PLO2, C3]

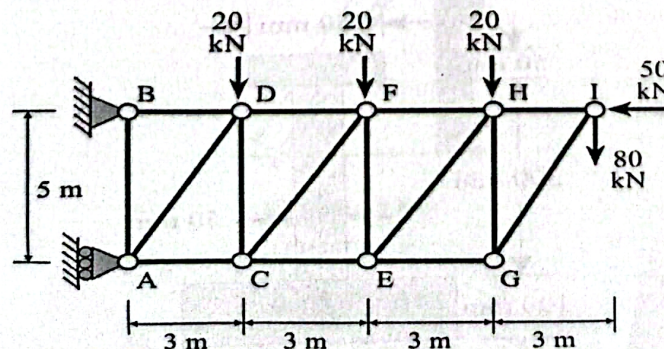


Figure: Q.8(b)